# Allan Paiz

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US Citizen | Relocating to Texas by January 2027

## Education

- B.S. in Computer Science from University of South Carolina — Expected – Dec 2026
  Minor in Mathematics | GPA: 3.7 / 4.0

## Relevant Work History

*Research Assistant at University of South Carolina, SC* — *Jan 2026 – Present*

- Worked under the direction of the Dynamical Systems and AI Research Group, assisting with hardware and simulation experiments involving autonomous mobile robots and control research.
- Developed and debugged robotics software in C++ and Python on Linux using ROS 2, integrating autonomy algorithms, physical robotics hardware, and Gazebo simulations.
- Wrote Python and Bash automation scripts for setup, execution, and data collection, creating repeatable workflows for multi agent path finding and autonomous navigation testing.
- Used Linux to collect and log large volumes of sensor, spatial, and network data while conducting live test integration, debugging, troubleshooting, and resolving software and hardware issues.

*Researcher at NSF CISE Research Experience for Undergraduates, SC* — *May 2026 – Aug 2026*

- Conducted robotics research using Python, ROS 2, computer vision, and sensor data to help autonomous mobile robots detect and navigate their environment. Prototyped computer vision and machine learning algorithms using experimental data collected from robotic hardware and sensors.
- Developed Python data processing and sensor fusion pipelines to transform and synchronize camera and LiDAR sensor data for environmental perception by applying an ETL process to raw sensor data extracting measurements and features and transforming them into structured outputs.
- Authored technical documentation communicating introductions to hardware, algorithms, mathematical foundations, and software architecture to technical and research audiences.

## Projects

*Event-Triggered Optimal Control for Obstacle-Avoidant Navigation* — *Paper*

- Collaborated with a PhD candidate and coauthored a research preprint on event triggered optimal control for autonomous mobile robot navigation.
- Formulated obstacle avoidant autonomous path planning as a constrained nonlinear optimization problem, applying algorithm development and mathematical modeling.
- Used Python and Bash on Linux systems to run algorithm simulations and collect experimental data for downstream control theory analysis.

*Defensive Stopping Power* *2024 NFL Big Data Bowl Finalist*

- Built a machine learning pipeline in Python, transforming raw tracking data into engineered spatial, velocity, and contextual features for modeling and statistical analysis.
- Developed a neural network to estimate contact probabilities using Python, Pandas, NumPy, and Keras, combining data processing, feature engineering, predictive modeling, and visualization.
- Translated model output into a new analytical KPI and communicated results through visualizations, rankings, and technical reporting for player scouting and performance analysis.

*Wash Your Rice* *Portfolio*

- Developed C++ game software using Unreal Engine 5 APIs, implementing user input, UI state management, simulated file navigation, dynamic screens, and progression logic.
- Worked through a video game development lifecycle using GitHub for collaboration, contributing to planning, design, development, automated testing, and scheduled iterative releases.
- Applied software architecture, API integration, object-oriented programming, version control, testing, and CI/CD practices while merging developed features into stable releases.

**Other Work History**

*Advanced Data Collector at Pro Football Focus, Remote* *May 2024 – Present*

- Worked independently managing high volume NFL and NCAA data, requiring time management, analytical judgment, attention to detail, and consistent delivery of accurate data.
- Performed data collection, validation, quality assurance, and error detection across multiple data categories, applying data quality controls to identify discrepancies and maintain dataset reliability.

*US Marine at United States Marine Corps, CA* *May 2018 – Oct 2022*

- Held a Secret Clearance, led, and managed small teams in mission critical, high pressure technical environments, applying problem solving, prioritization, and decision-making skills.
- Used digital reconnaissance and GIS systems to collect, analyze, and communicate geospatial data for engineering and operational planning.

**Skills**

***Languages:*** C++, Python, Java, Bash; Pandas, NumPy, Plotly, Matplotlib, Scikit-learn, Junit
***Tools:*** Linux, Git, GitHub, LaTeX, Excel, PowerPoint, Neovim, Photoshop, Premiere Pro, Unreal Engine 5

***Relevant Coursework:*** Software Engineering, Analysis of Experimental Data in Python, Optimization, Math Foundations of ML/Data Science, Artificial Intelligence, Computer Architecture, Operating Systems, Linear Algebra, Technical Writing, Elementary Number Theory, Introduction to Computer Networks